

24UAMP10	ML OPS	L/T/P/C 2/0/2/3
Nature of Course: Lab Integrated Theory		PEC
Prerequisite(s): Basics of Python Programming		
Course Objectives: Enable the students to		
○	Analyze the principles of MLOps, version control systems, and the MLOps lifecycle for managing machine learning workflows.	
○	Apply experiment tracking, feature stores, and metadata management techniques to improve reproducibility and scalability of ML systems.	
○	Design and implement CI/CD pipelines and orchestration workflows for automated machine learning model development and deployment.	
○	Develop and evaluate model serving and monitoring systems to ensure reliability, performance, and drift detection in production ML systems.	
○	Assess and apply best practices in MLOps including reliability engineering, secure deployment, compliance, and emerging trends such as LLMOps and edge AI.	
Course Contents:		
UNIT I	FOUNDATIONS OF MLOPS & VERSION CONTROL	CO1 6
Introduction to MLOps - Need for MLOps - DevOps vs MLOps -MLOps lifecycle stages - Version Control in MLOps - Code versioning - Data versioning - Model versioning - Environment and Dependency Management - Use of virtual environments - Basics of Docker and MLOps pipelines.		
UNIT II	EXPERIMENT TRACKING AND FEATURE STORES	CO2 6
Experiment Tracking:Logging metrics, parameters, artifacts - Use of MLflow Tracking - Comparing experiments, understanding overfitting, underfitting through logs- Feature Store Fundamentals: Feature engineering vs feature selection -Online vs offline stores, Tools: Feast, Tecton, Hopsworks - Metadata Management: Model lineage, tracking data/model dependencies Introduction to ML metadata stores (e.g., ML Metadata, Kube Flow Metadata		
UNIT III	CI/CD & ORCHESTRATION FOR ML PIPELINES	CO3 6
CI/CD in ML: Concepts of continuous integration, testing ML code - CI/CD of ML models - GitHub Actions, Jenkins for pipeline automation - Pipeline Orchestration: , Directed Acyclic Graph (DAG) concepts - Workflow management using Apache Airflow, Kubeflow Pipelines - Containerization and Deployment: Docker for containerizing ML models - Introduction to Kubernetes and container orchestration - TFX (TensorFlow Extended) pipeline - Deployment Strategies: Canary release, blue-green deployment - A/B testing for model rollouts		
UNIT IV	MODEL SERVING AND MONITORING	CO4 6
Model Serving Frameworks: Model inference APIs using FastAPI, Flask - Serving ML models via TensorFlow Serving, TorchServe, BentoML - Model Monitoring: Types of monitoring: performance, drift, bias, latency Tools: EvidentlyAI,Prometheus, Grafana, Fiddler, Arize AI - Setting up real-time dashboards - Detecting and responding to concept/data drift Retraining Workflows: Triggering retraining pipelines based on performance Human-in-the-loop feedback systems		
UNIT V	BEST PRACTICES AND EMERGING TRENDS	CO5 6
ML Reliability and SRE Principles:Failover strategies, rollback mechanisms - ML incident response and alerting - Principles from SRE - LLMOps - Responsible MLOps : - Compliance and Security in ML Systems: Secure MLOps Regulations - Model validation - Future Trends: ML and edge computing - Multi Cloud MLOps.		
Theory Hours:		30

PRACTICAL EXERCISES		
1.	Implement a solution using DVC and MLflow to track data, code, model versions, and experiment metrics, ensuring full reproducibility of the ML pipeline.	CO1
2.	Use an open-source feature store (Feast) to ingest, manage, and retrieve consistent offline and online features for training and inference.	CO2
3.	Containerize the model using Docker and deploy it in a Kubernetes cluster to ensure consistent, scalable, and portable deployment.	CO3
4.	Design a CI/CD pipeline using GitHub Actions and Airflow that automates model testing, training, validation, and deployment on each code commit.	CO3

5.	Develop and deploy a REST API using FastAPI (or Flask) to serve predictions from your model, and test it using realistic inputs.	CO4
6.	Implement monitoring and drift detection using EvidentlyAI, Prometheus, and Grafana to detect data and concept drift in real time.	CO4
7.	Simulate performance degradation, then build an automated retraining pipeline using Airflow that retrains and replaces the model, with notification alerts.	CO5
8.	Open Ended Experiment	CO1CO5
Practical Hours:		30
Total Hours:		60

COURSE OUTCOMES	
At the end of the course students will be able to	
CO1	Analyze the foundations of MLOps including DevOps vs MLOps, version control, environment management, and Docker-based pipelines.
CO2	Apply experiment tracking, feature store concepts, and metadata management using tools like MLflow, Feast, and ML metadata systems
CO3	Design CI/CD pipelines and ML workflow orchestration systems using tools such as GitHub Actions, Jenkins, Airflow, Kubeflow, and Docker/Kubernetes.
CO4	Develop and evaluate ML model serving and monitoring systems using APIs, monitoring tools, and drift detection technique
CO5	Assess and implement reliable, secure, and scalable MLOps systems following SRE principles, compliance standards, and emerging trends.

Text Books:	
1	Andrew McMahon, "Machine Learning Engineering with Python: Manage the Lifecycle of Machine Learning Models Using MLOps with Practical Examples", Apress, 2023
2	Noah Gift, Alfredo Deza, "Practical MLOps", O'Reilly, 2022
3	Chip Huyen, "Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications", O'Reilly, 2022
Reference Books:	
1	Cathy Chen et al., "Reliable Machine Learning: Applying SRE Principles to ML in Production", O'Reilly, 2022

2	Mark Treveil & Dataiku Team, "Introducing MLOps: How to Scale Machine Learning in the Enterprise", O'Reilly,2022													
3	Emmanuel Raj, "Engineering MLOps", Packt Publishing, 2023													
Online Resources:														
1	https://www.coursera.org/specializations/mlops-machine-learning-duke													
2	https://github.com/DataTalksClub/mlops-zoomcamp													
CO/ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	-	3	-	-	2	2	-	2	2	3	-
CO2	2	3	2	-	3	-	-	2	2	-	2	2	3	2
CO3	2	3	3	-	3	-	-	2	2	-	2	2	3	2
CO4	2	3	3	-	3	-	-	2	2	-	2	2	3	2
CO5	2	3	3	-	3	2	2	2	2	-	3	2	3	3

Correlation levels -, 2 and 3 are as defined below: - - Slight 2 Moderate 3 - Substantial (High)

HoD

Dean - Academics

Principal